

Advanced (Habanero)

Q1. Simplify $2\sqrt{3} + 5\sqrt{3}$

- (A) $7\sqrt{3}$
- (B) $7\sqrt{6}$
- (C) $1\sqrt{3}$
- (D) $6\sqrt{3}$
- (E) $12\sqrt{3}$

Q2. Simplify $\sqrt{12} - \sqrt{3}$

- (A) $\sqrt{9}$
- (B) $\sqrt{15}$
- (C) $2\sqrt{3}$
- (D) $\sqrt{12} - \sqrt{3}$
- (E) $\sqrt{3} - \sqrt{4}$

Q3. Simplify $(2\sqrt{2})(6\sqrt{2})$

- (A) $12\sqrt{4}$
- (B) 24
- (C) 12
- (D) $4\sqrt{2}$
- (E) 8

Q4. Simplify $\frac{\sqrt{20}}{\sqrt{5}}$

- (A) $\sqrt{20}$
- (B) 2
- (C) $\sqrt{4}$
- (D) $\sqrt{5}$
- (E) $\sqrt{2}$

Q5. Simplify $3\sqrt{2} + 2\sqrt{5}$

- (A) $5\sqrt{7}$
- (B) $3\sqrt{2} + 2\sqrt{5}$
- (C) $\sqrt{10}$
- (D) $\sqrt{14}$
- (E) $5\sqrt{2}$

Q6. Simplify $(\sqrt{3} + \sqrt{2})^2$

- (A) $7 + 2\sqrt{6}$
- (B) $5 + 2\sqrt{6}$
- (C) $3 + 2 + 2\sqrt{6}$
- (D) $\sqrt{3} + \sqrt{2}$
- (E) $2\sqrt{3}$

Q7. Simplify $(\sqrt{8})(\sqrt{2})$

- (A) 4
- (B) 8
- (C) $\sqrt{16}$
- (D) $\sqrt{4}$
- (E) $2\sqrt{2}$

Q8. Simplify $(\sqrt{18})^2$

- (A) 18
- (B) 9
- (C) 6
- (D) $3\sqrt{2}$
- (E) $4\sqrt{2}$

Open Ended Questions (FRQ)

Q9. Simplify $\frac{(\sqrt{12} + \sqrt{3})}{(\sqrt{4} - \sqrt{3})}$

Q10. Multiply $(2\sqrt{6})(3\sqrt{2})$ and simplify your answer

Chef's Tips

- Remind students to check their work and explain their reasoning
- Suggest students use a calculator to verify their answers
- Encourage students to think critically and apply different problem-solving strategies